CO2xN: Porewater Sulfide

Month Year Samples

2026-01-29

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
  #And Will Write a Rerun File

###things that need to be changed

sample_month = "July"
sample_year = "2025"
Run_Date = as.Date("07/14/2025", format = "%m/%d/%Y")
Run_by = "Melanie Giessner"#Instrument user
Script_run_by ="Brenden Blakley"#Code user

Run_notes=""#any notes from run

#Import Plate Data

plates<- c("Plate1","Plate2","Plate3","Plate4","Plate5","Plate6","Plate7", "Plate8", "Plate9", "Plate10",

Raw_plates <- c("Raw Data/20250714_LTREB_H2S_Plate1.csv",
                "Raw Data/20250714_LTREB_H2S_Plate2.csv",
                "Raw Data/20250714_LTREB_H2S_Plate3.csv",
                "Raw Data/20250714_LTREB_H2S_Plate4.csv",
                "Raw Data/20250714_LTREB_H2S_Plate5.csv",
                "Raw Data/20250714_LTREB_H2S_Plate6.csv",
                "Raw Data/20250715_LTREB_H2S_Plate7.csv",
                "Raw Data/20250715_LTREB_H2S_Plate8.csv",
                "Raw Data/20250715_LTREB_H2S_Plate9.csv",
                "Raw Data/20250715_LTREB_H2S_Plate10.csv",
                "Raw Data/20250716_LTREB_H2S_reruns.csv")

#Import Run Date for Each Plate for Std curve qaqc
Run_dates <- tibble(
    Plate = c("Plate1","Plate2","Plate3","Plate4","Plate5","Plate6","Plate7","Plate8","Plate9","Plate10",
    Run_Date = c("07/14/2025","07/14/2025","07/14/2025","07/14/2025","07/14/2025","07/14/2025","07/15/2025
  )

  Run_dates <- Run_dates %>% mutate(Run_Date = as.Date(Run_Date, format = "%m/%d/%Y"))


# Define the file path for QAQC log file - NO Need to change just check year
```

```r
log_path <- "Processed Data/LTREB_2M_H2S_QAQC_2025.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_CO2xN_Porewater_H2S_202507.csv"
rerun_path <- "Processed Data/GCReW_CO2xN_Porewater_H2S_202507_reruns.csv"
```

##Read in Constants

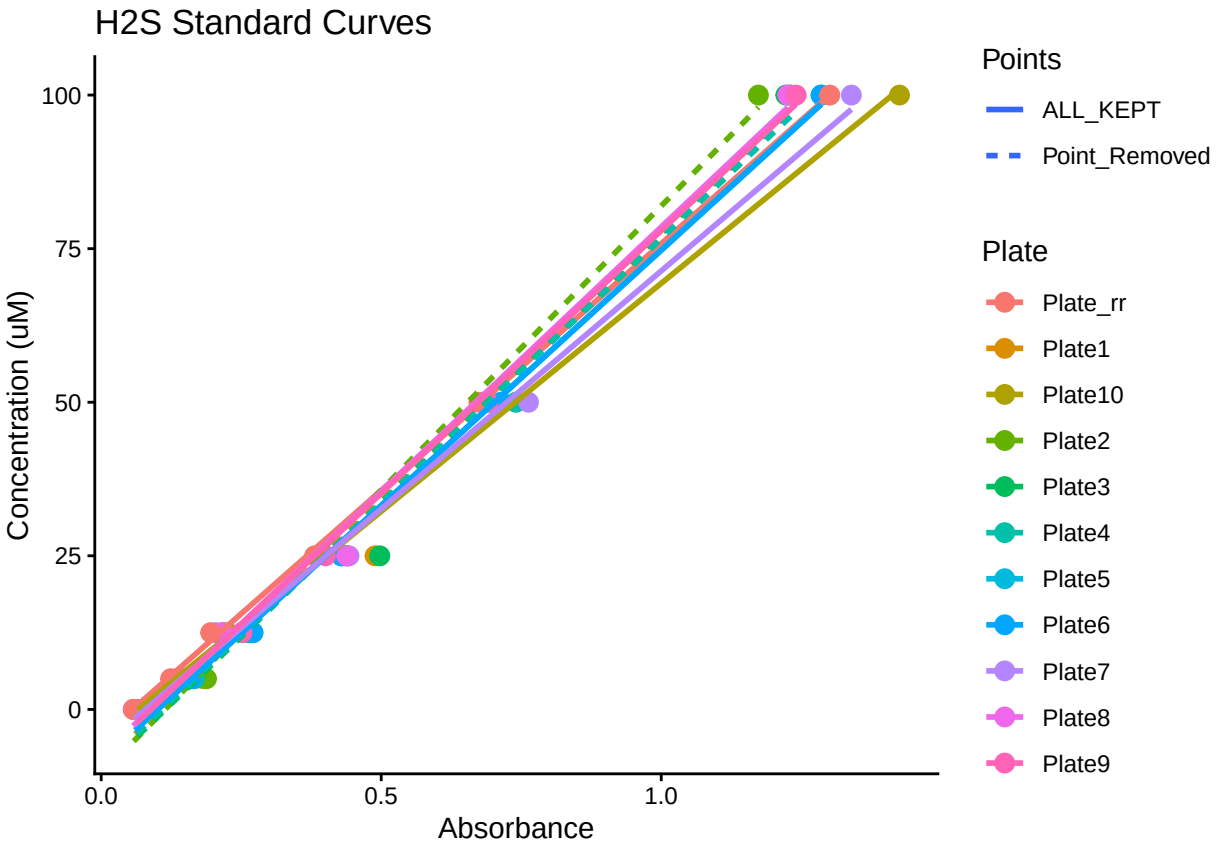
##Read in metadata and create similar sample IDs for matching to samples

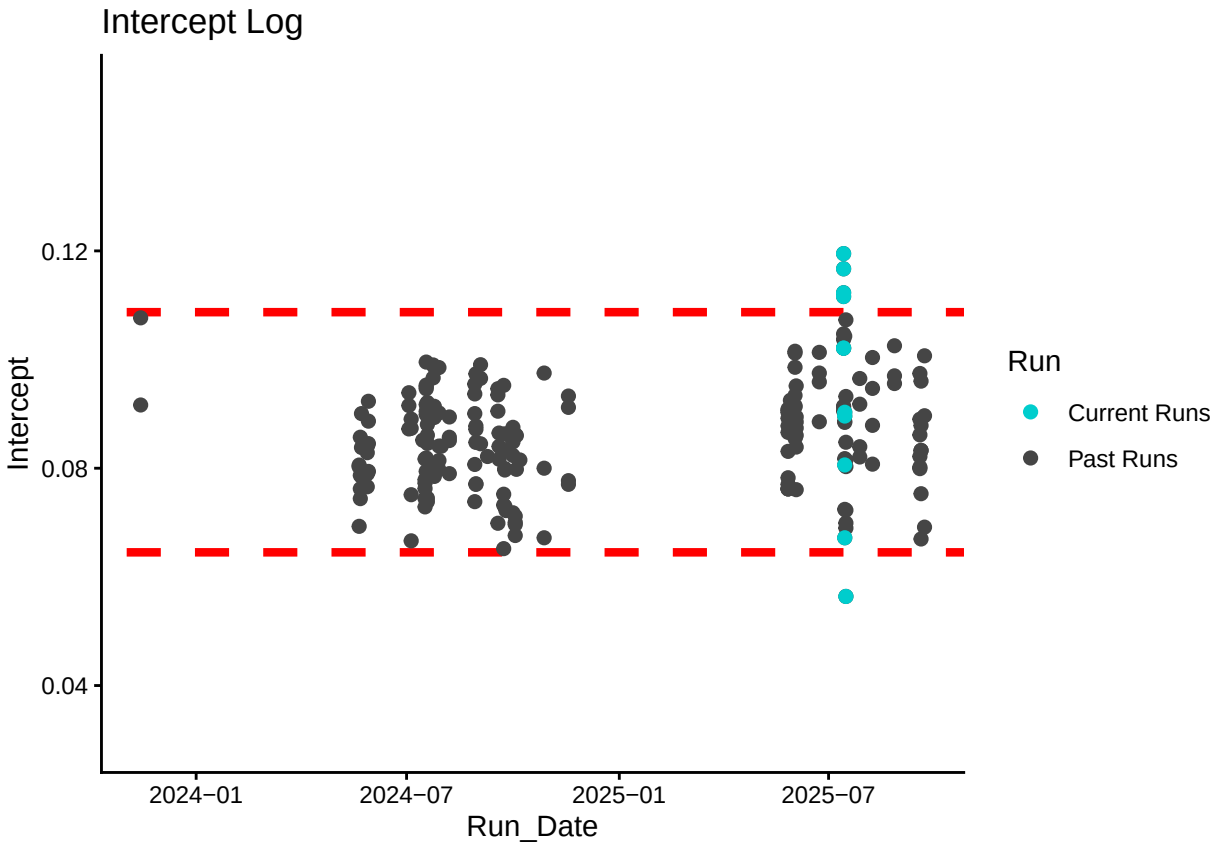
##Read in data

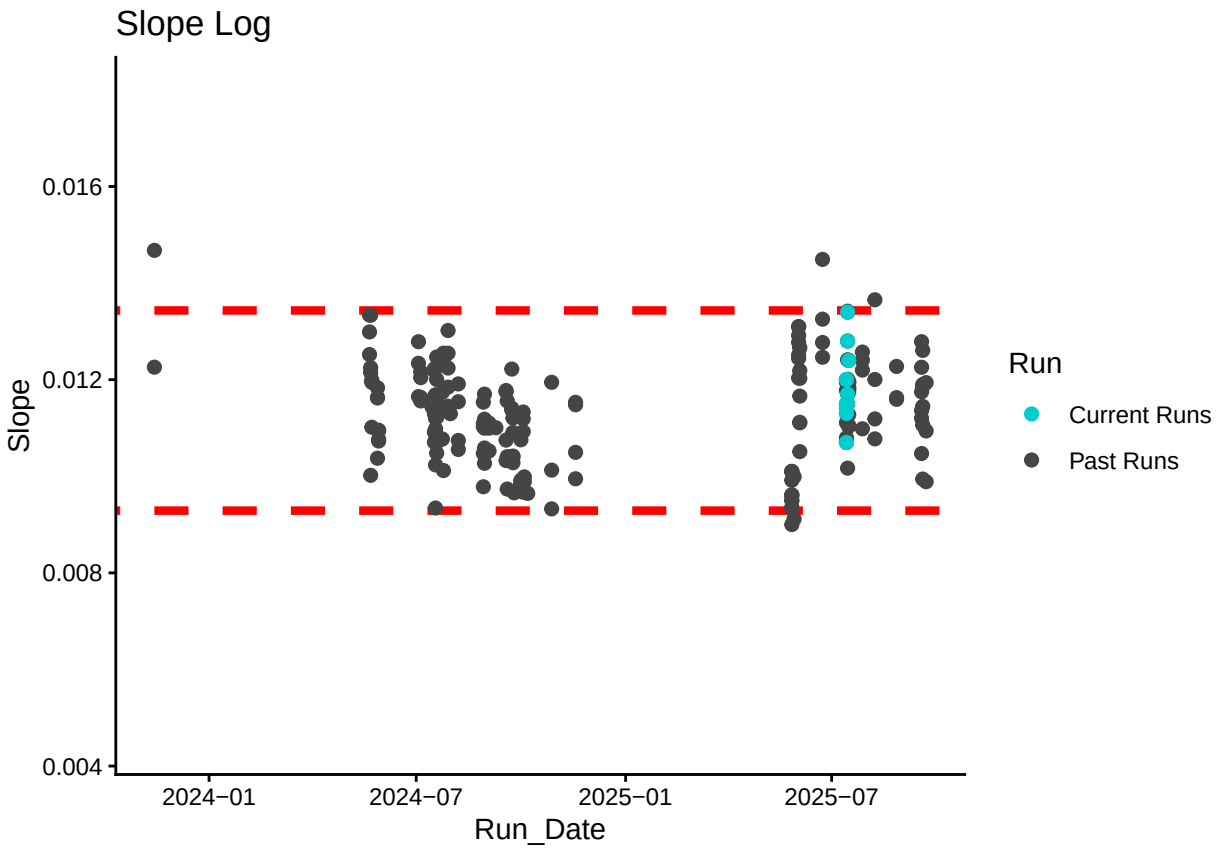
##Remove High CV Stds

##Check R2 & Remove Standards if Necessary

Plot Standard Curves







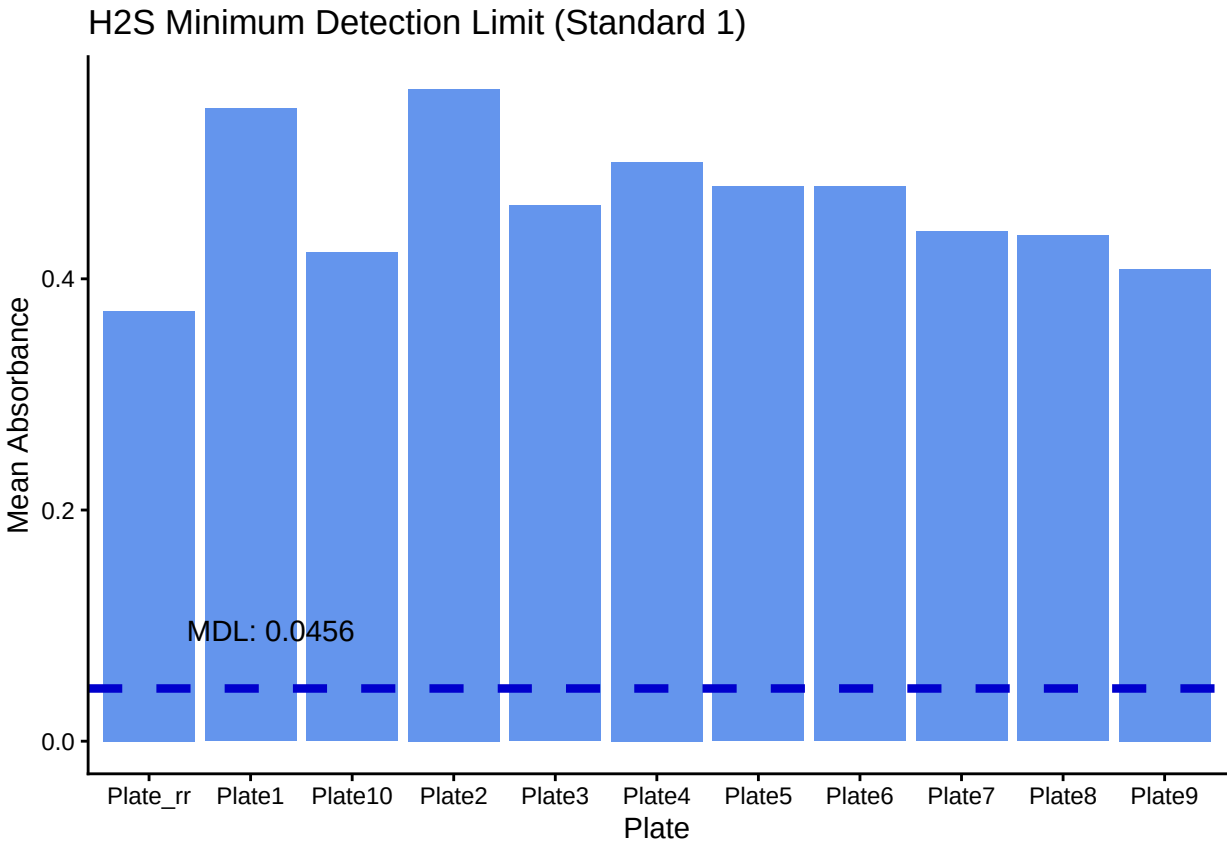
Return Flags For Standard Curve

```
##  [1] "CV Good-Plate1"   "CV Good-Plate10"  "CV Good-Plate2"   "CV Good-Plate3"
##  [5] "CV Good-Plate4"   "CV Good-Plate5"   "CV Good-Plate6"   "CV Good-Plate7"
##  [9] "CV Good-Plate8"   "CV Good-Plate9"   "CV Good-Plate_rr"
```

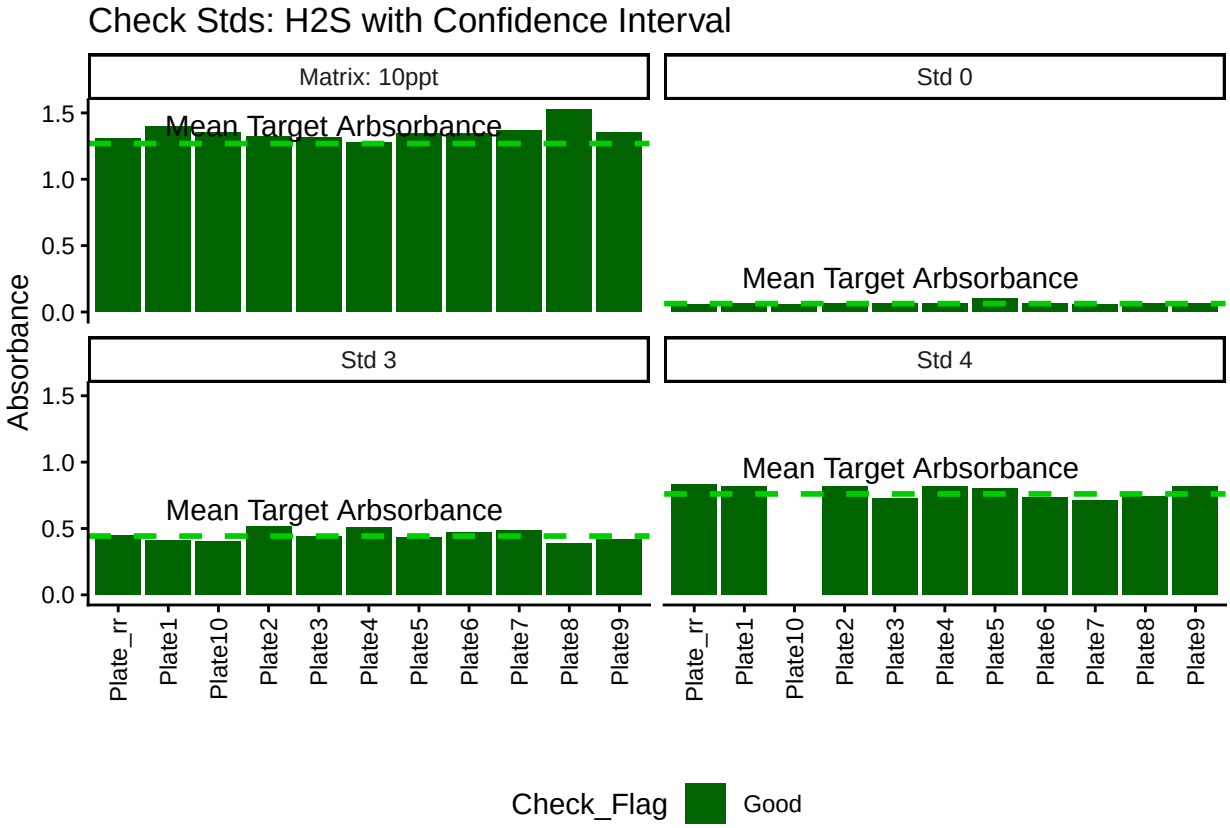
```
##  [1] "R2 Bad-Plate1"    "R2 Good-Plate10"  "R2 Good-Plate2"   "R2 Bad-Plate3"
##  [5] "R2 Good-Plate4"   "R2 Good-Plate5"   "R2 Good-Plate6"   "R2 Good-Plate7"
##  [9] "R2 Good-Plate8"   "R2 Good-Plate9"   "R2 Good-Plate_rr"
```

```
##  [1] "Curve_Removed-Plate1"   "Curve_Removed-Plate2"   "Curve_Removed-Plate3"
##  [4] "Curve_Removed-Plate4"   "Curve_Kept-Plate5"      "Curve_Kept-Plate6"
##  [7] "Curve_Kept-Plate7"      "Curve_Kept-Plate8"      "Curve_Kept-Plate9"
## [10] "Curve_Kept-Plate10"     "Curve_Removed-Plate_rr"
```

Method Detection Limit



Compare Check Standards to Standards



Display Any Check Flags

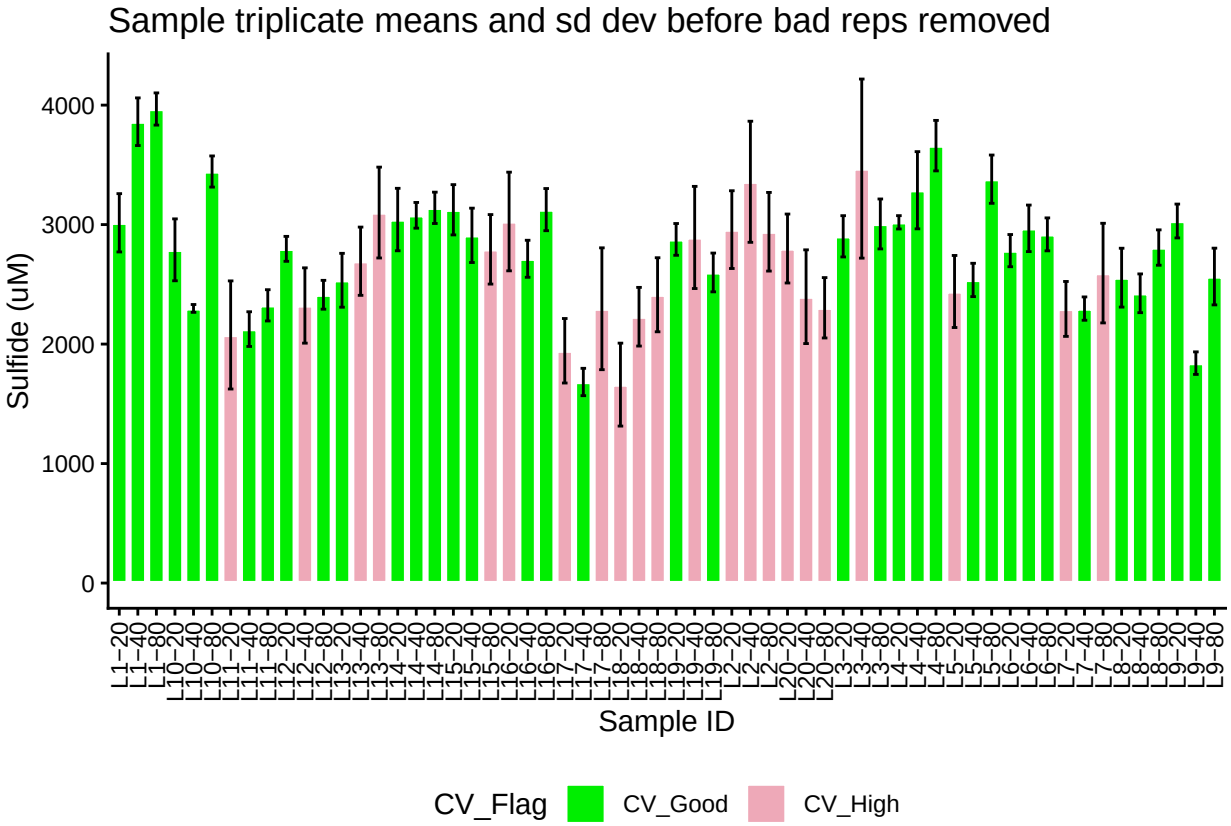
```
## [1] "All Checks Good"
```

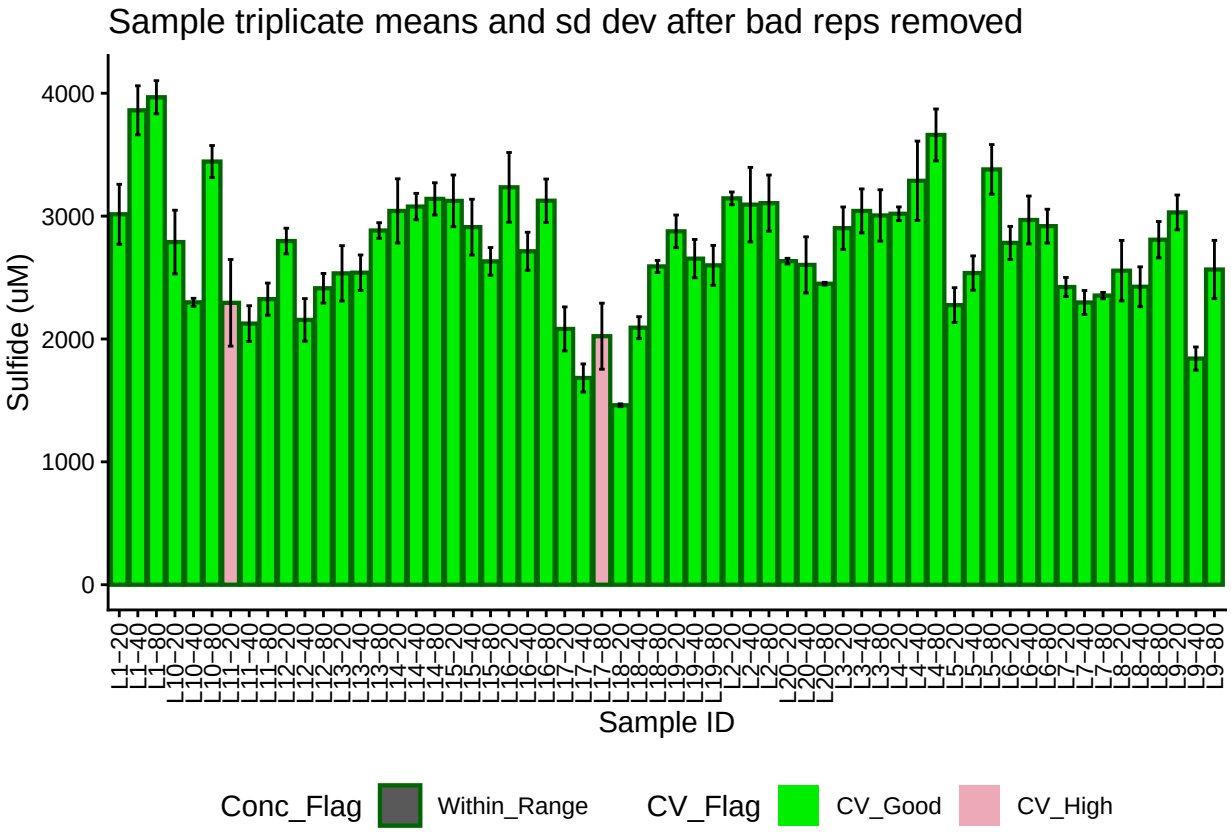
```
## [1] "Std0Chk_CV_High-Plate5"
```

##Calculate Sulfide Concentrations & Add Concentration Flags

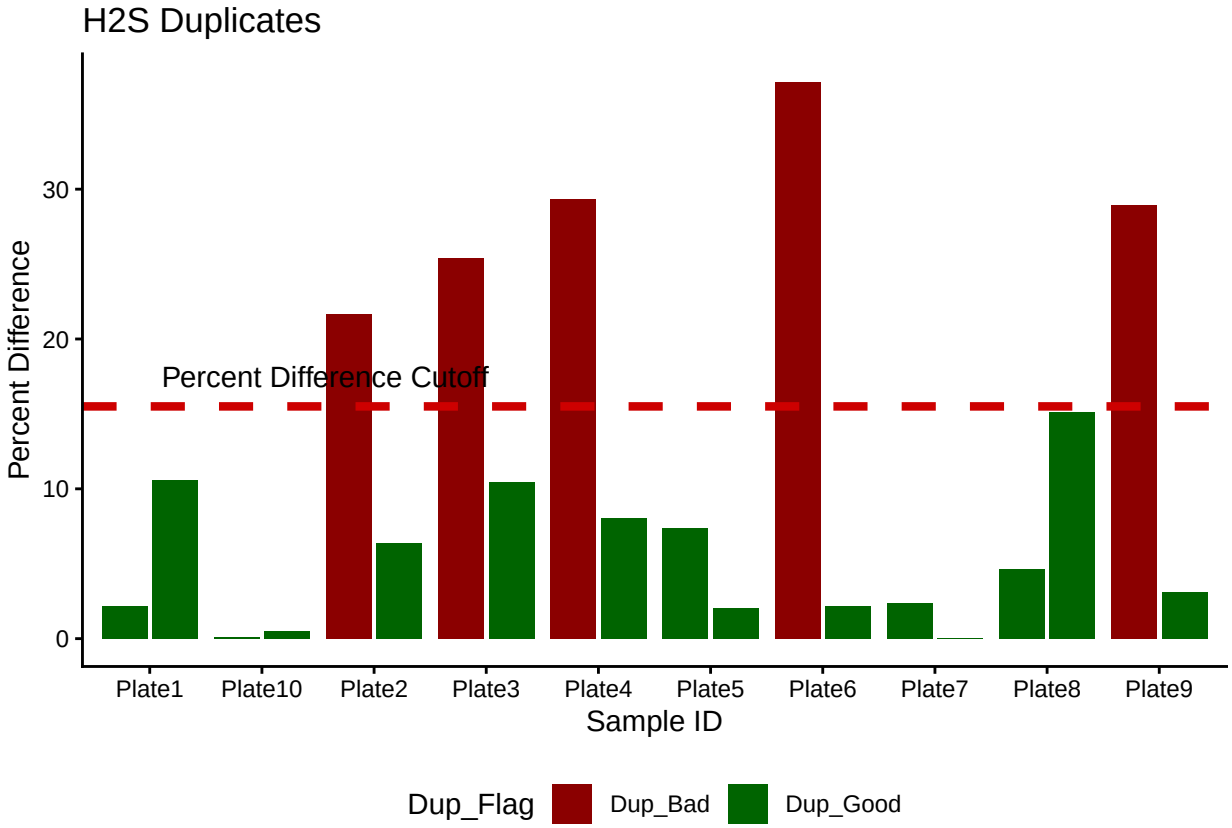
##Remove High CV Samples and Average

Plot Samples





Duplicate QAQC



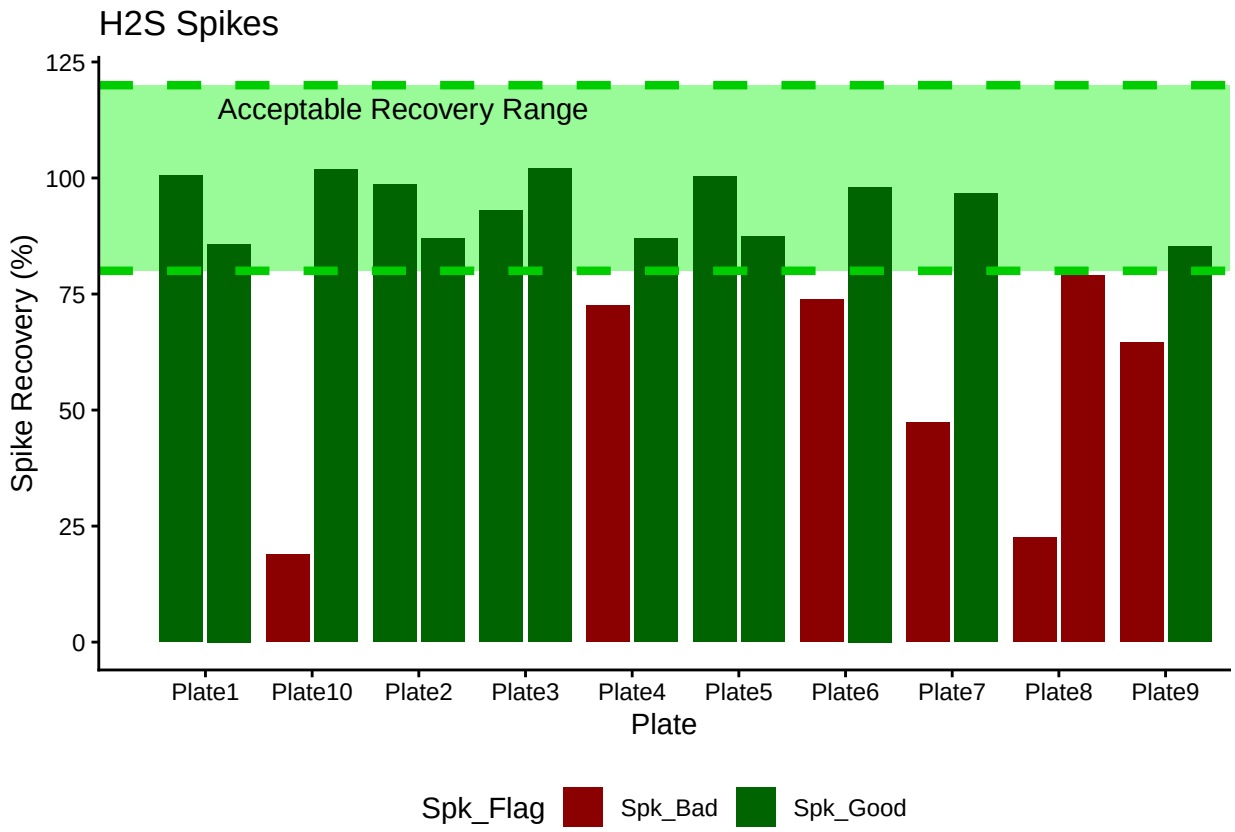
Display Dup Flags

```
## [1] ">60% Dups Pass"
```

```
##  [1] "Dup_Good-Plate1"  "Dup_Bad-Plate2"   "Dup_Bad-Plate3"   "Dup_Bad-Plate4"
##  [5] "Dup_Good-Plate5"  "Dup_Bad-Plate6"   "Dup_Good-Plate7"  "Dup_Good-Plate8"
##  [9] "Dup_Bad-Plate9"   "Dup_Good-Plate10"
```

Spike QAQC

```
## Warning in left_join(., spk_flags_1, by = join_by(Spk_ID, Plate)): Detected an unexpected many-to-many r
## i Row 38 of ‘x‘ matches multiple rows in ‘y‘.
## i Row 2 of ‘y‘ matches multiple rows in ‘x‘.
## i If a many-to-many relationship is expected, set ‘relationship =
##   "many-to-many"‘ to silence this warning.
```



Display Spike Flags

```
## [1] "Spikes Paired"
```

```
##  [1] "Spk_Good-Plate1" "Spk_Good-Plate2" "Spk_Good-Plate3" "Spk_Bad-Plate4"
##  [5] "Spk_Good-Plate5" "Spk_Bad-Plate6"  "Spk_Bad-Plate7"  "Spk_Bad-Plate8"
##  [9] "Spk_Bad-Plate9"  "Spk_Bad-Plate10"
```

```
## [1] "<60% of Spikes Pass, Rerun"
```

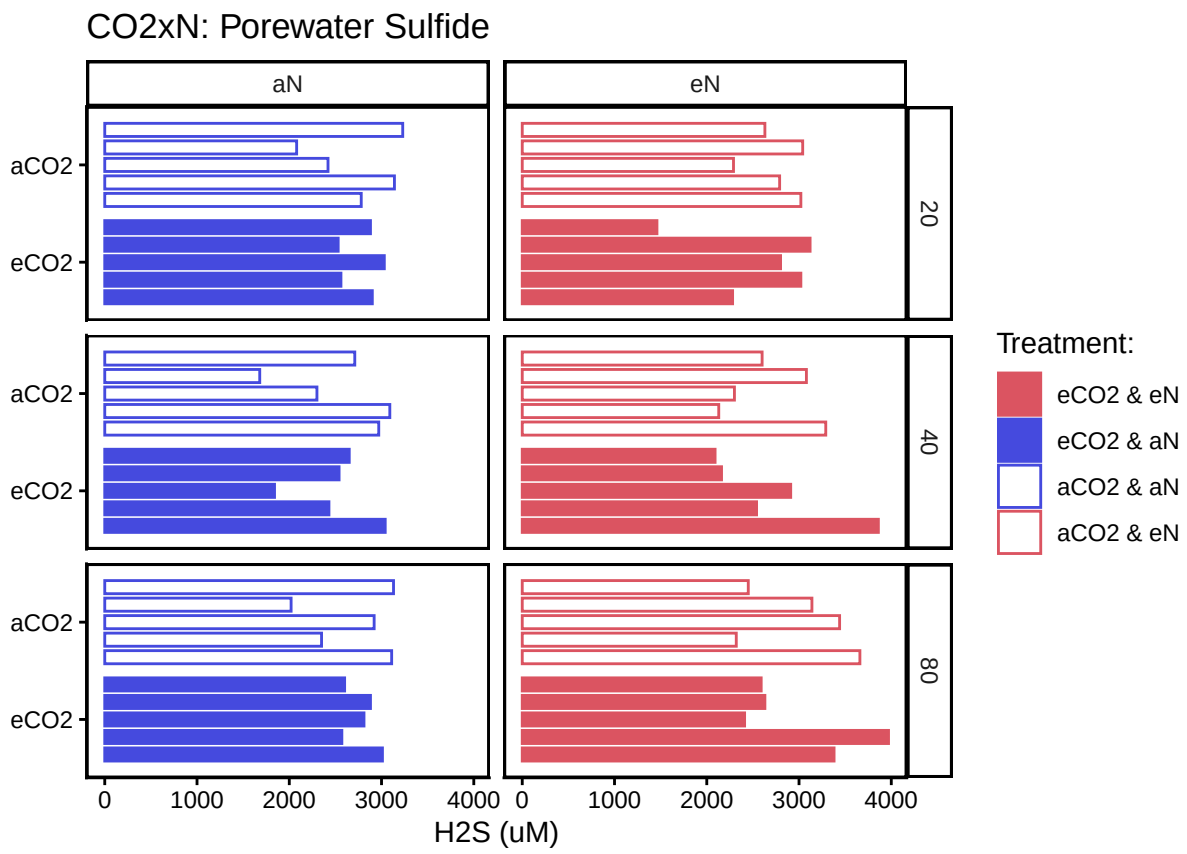
Merge samples with Metadata & check all samples are present

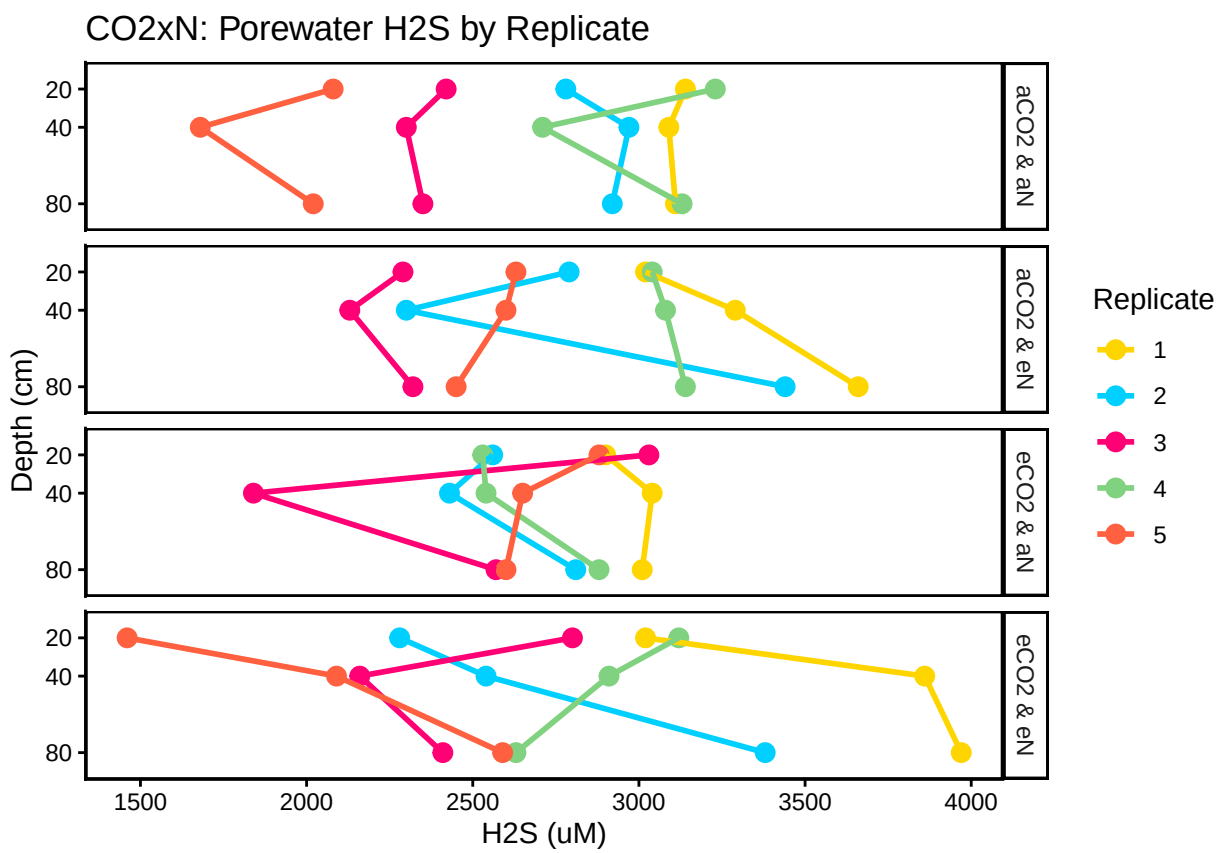
```
## All sample IDs are present in data
```

##Add Flags

##Organize Data

Visualize Data





##Export Data

END